

Technical Document #99: Database Systems - Comprehensive Overview

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Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Database caching stores frequently accessed data in memory. Redis and Memcached are popular caching solutions.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Database connection pooling reuses database connections to reduce overhead.
- ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.
- Database monitoring tracks performance metrics and identifies bottlenecks.